

Product Requirements Document (PRD)

Feature: Typing Behavior Analytics

Product: BLUEBOOK

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Status: Implemented / Case Study Feature

1. Executive Summary

BLUEBOOK aims to modernize assignment submission workflows while promoting digital literacy and authentic student participation.

As AI-powered writing tools and browser extensions become increasingly accessible, students can complete assignments with minimal engagement by relying entirely on generated content or direct copy-paste methods.

The Typing Behavior Analytics feature is designed to encourage active participation during assignment creation by analyzing writing behavior patterns during the submission process. The feature focuses on how content is produced rather than evaluating the content itself.

Its objective is to support learning through participation while maintaining a digital-first educational experience.

2. Problem Statement

Traditional handwritten assignments naturally require active student involvement. However, digital assignments introduce a new challenge.

Students can submit content generated by AI tools or copied from external sources without engaging meaningfully with the assignment.

This creates several issues:

- Reduced participation in the learning process.
- Increased dependence on AI-generated content.
- Difficulty distinguishing actively written work from passive submissions.
- Reduced educational value of assignments.

Current solutions such as plagiarism detection primarily evaluate content after submission. They do not provide insight into the writing process itself.

3. Background

During research and observation of student assignment workflows, two trends became apparent:

Trend 1: Digital Adoption

Educational institutions are increasingly moving toward digital submission systems.

Trend 2: AI-Assisted Content Generation

Students now have access to tools capable of generating entire assignments within seconds.

While AI can be beneficial for learning, excessive reliance may reduce student engagement and critical thinking.

This created an opportunity to design a feature that promotes participation rather than simply detecting misconduct.

4. Goals

Primary Goals

- Encourage active participation during assignment writing.
- Promote authentic engagement with assignments.
- Complement digital assignment workflows.
- Support educators with additional participation insights.

Secondary Goals

- Promote responsible AI usage.
- Increase awareness of writing habits.
- Improve overall assignment quality.
- Create a foundation for future learning analytics.

5. Success Metrics

The feature will be considered successful if it achieves the following:

Student Metrics

- Increased time spent actively writing assignments.
- Reduced dependence on direct content insertion.
- Improved engagement with assignment tasks.

Platform Metrics

- Percentage of submissions containing active typing behavior.

- Number of assignments completed through direct writing workflows.
- Adoption rate of digital assignment submissions.

Teacher Metrics

- Increased confidence in assignment authenticity.
- Improved visibility into submission behaviors.

6. User Personas

Student Persona

Description

College student completing digital assignments.

Needs

- Simple assignment submission.
- Flexible writing environment.
- Minimal friction during workflow.

Pain Points

- Temptation to rely entirely on AI-generated responses.
- Limited accountability in digital submission systems.

Teacher Persona

Description

Faculty member managing assignments and evaluating submissions.

Needs

- Better visibility into student participation.
- Efficient assignment management.

Pain Points

- Difficulty identifying passive submission behavior.
- Lack of insight into assignment creation processes.

7. User Stories

Student

As a student, I want to complete assignments digitally so that I can avoid handwritten submissions.

Student

As a student, I want a simple writing experience so that I can focus on completing my work.

Teacher

As a teacher, I want visibility into student participation patterns so that I can better understand engagement levels.

Teacher

As a teacher, I want assignment workflows to remain fair and authentic.

8. Proposed Solution

Introduce a Typing Behavior Analytics system that observes writing activity during assignment creation.

The feature captures writing-related behavioral signals that indicate active engagement with the submission process.

Rather than analyzing assignment content, the feature evaluates participation characteristics.

This allows BLUEBOOK to focus on learning behavior rather than content judgment.

9. Functional Requirements

Assignment Writing Session

The system shall track writing sessions while students are composing assignments.

Typing Activity Monitoring

The system shall record typing activity patterns during assignment creation.

Submission Analysis

The system shall process writing behavior data before assignment submission.

Participation Indicators

The system shall generate participation-related metrics for each submission.

Teacher Visibility

The system shall provide participation insights alongside assignment submissions

10. Non-Functional Requirements

Performance

The feature shall not negatively impact writing experience.

Privacy

Collected data shall be limited to writing behavior signals relevant to participation analysis.

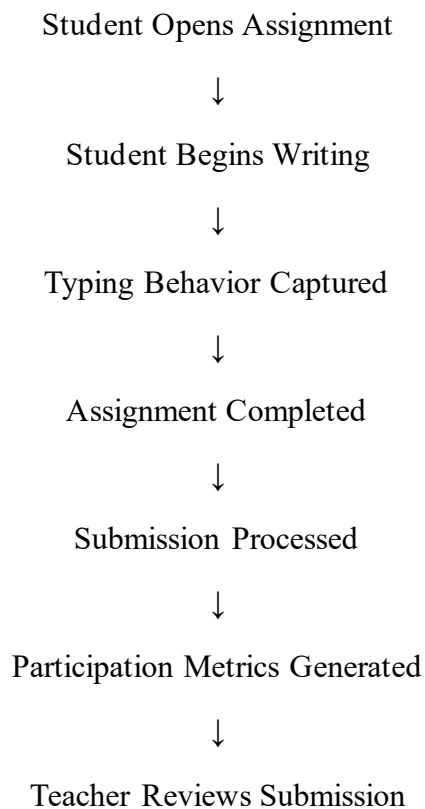
Scalability

The feature shall support increasing numbers of assignments and users.

Reliability

Behavior tracking shall function consistently throughout writing sessions.

11. User Flow



12. Risks

False Interpretation

Some students naturally type differently from others.

Mitigation

Use participation indicators as supplementary information rather than definitive judgments.

Privacy Concerns

Students may have concerns regarding behavioral data collection.

Mitigation

Clearly communicate the purpose and scope of data collection.

Technical Complexity

Behavior analysis requires additional processing and storage.

Mitigation

Implement efficient data collection and processing mechanisms.

13. Future Enhancements

AI Writing Insights

Provide students with personalized writing improvement suggestions.

Learning Analytics Dashboard

Offer deeper engagement insights to educators.

Participation Trends

Track engagement patterns across multiple assignments.

Early Intervention Alerts

Identify students who may need additional academic support.

14. Expected Impact

For Students:

- Increased participation.
- Greater engagement with assignments.
- Improved digital literacy.

For Teachers:

- Better visibility into learning behaviors.
- Additional context during assignment evaluation.

For Institutions:

- More effective digital assignment workflows.
- Stronger alignment between technology and learning objectives.

15. Key Product Principle

The purpose of Typing Behavior Analytics is not to police students.

The purpose is to encourage active participation, support authentic learning experiences, and ensure that technology enhances education rather than replacing the learning process itself.